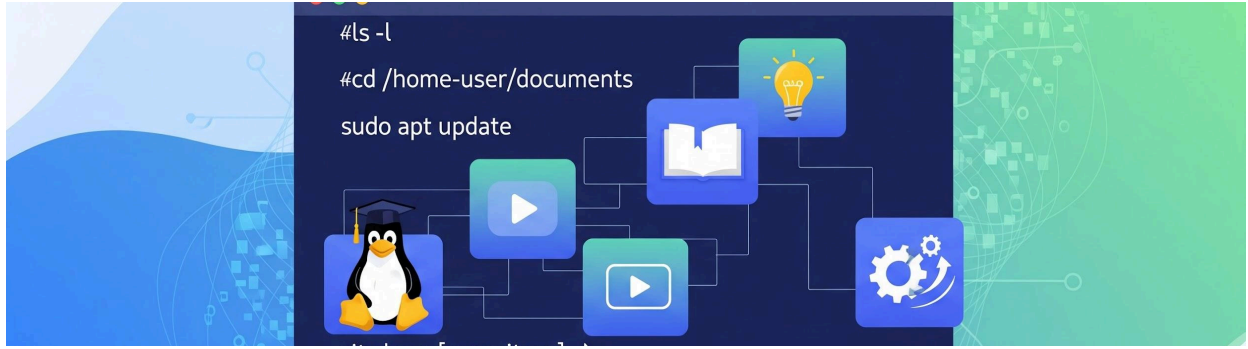




LINUX OS

Introduction:

- Linux is a free and open-source operating system with high security.
- Linux is a multi-user operating system.

OPERATING SYSTEM

- An Operating System (OS) is software that acts as an interface between computer hardware components and the user.
- Every computer system must have at least one operating system to run other programs.
- Applications like browsers, MS Office, Notepad, games, etc., need an environment to run and perform tasks.
- The OS helps users communicate with the computer without knowing the computer's language.

KERNEL

- The kernel manages the hardware components such as CPU, memory, and peripheral devices.
- The kernel is the lowest level of the operating system.

DAEMONS

- Daemons manage background services such as printing, sound, scheduling, etc.
- These services start during system boot or after user login.

SHELL

- The shell is an environment where users can run commands, programs, and shell scripts.
- It takes input from the user, executes it, and displays the output.
- A shell acts as an interface between the user and the operating system.

Shells can be:

Ø Graphical (GUI-based)

Ø Command-line based (CLI)

COMMAND

- A command is an instruction or request given to the operating system by a user.
- It tells the computer to perform a specific task.

TERMINAL

- A terminal is a text-based interface that allows users to interact with the operating system by typing commands and receiving output.
- Terminals are commonly used in Linux, Unix, macOS, and Windows systems.

LINUX OS DISTRIBUTIONS

- Linux distributions are modified versions of Linux released with different names based on user requirements.

Examples:

- i. RedHat
- ii. Ubuntu
- iii. Debian
- iv. CentOS
- v. Fedora
- vi. OpenSUSE
- vii. Kali Linux
- viii. Amazon Linux

ix. Rocky Linux

LINUX HISTORY

- In 1991, Linus Torvalds, a student at the University of Helsinki, Finland, started developing Linux.
- The Linux kernel is written in C language.
- Initially, the name was planned as Freax, but later it became Linux.
- In 1992, Linux was released under the GNU General Public License.

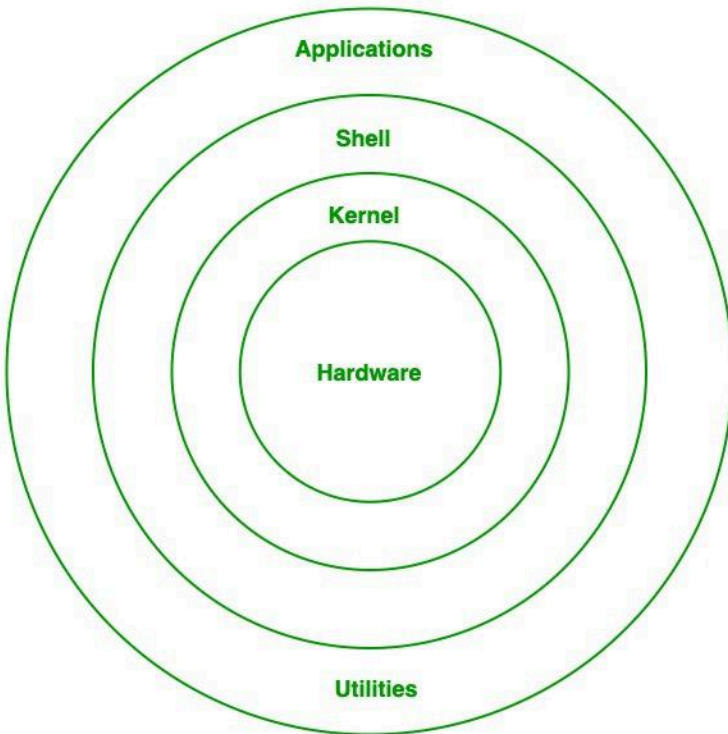
OPEN SOURCE

- Linux is distributed under an open-source license.
- Open source provides:
 - Freedom to use the software
 - Freedom to study and modify the code
 - Freedom to redistribute copies
 - Freedom to distribute modified versions

ARCHITECTURE

The Linux architecture consists of:

- Kernel
- Shell
- Commands
- Terminal



Introduction to Essential Linux Commands

Linux is a powerful and popular operating system, especially for servers, developers, and advanced users. To effectively use Linux, you interact with it through the **Command Line Interface (CLI)**, also known as the terminal or shell.

This document provides a detailed, yet simple, guide to essential Linux commands.

Basic Navigation and File Management

These commands help you move around the filesystem and handle files and directories (folders).

Command	Simple Language Explanation	Example
<code>pwd</code>	P rint W orking D irectory. Shows you the full path of the directory you are currently in.	<code>pwd</code>
<code>ls</code>	L ist. Shows you the files and folders inside your current directory.	<code>ls</code>
<code>ls -l</code>	A "long list" that shows detailed information (permissions, size, owner, date).	<code>ls -l</code>
<code>cd</code>	C hange D irectory. Used to move from one directory to another.	<code>cd Documents</code> (moves into the Documents folder)
<code>cd ..</code>	Moves you up one level (to the parent directory).	<code>cd ..</code>
<code>cd ~</code>	Moves you directly to your user's Home directory (the starting point).	<code>cd ~</code>
<code>mkdir</code>	M ake D irectory. Creates a new folder.	<code>mkdir new_project</code>
<code>touch</code>	Creates a brand-new, empty file.	<code>touch notes.txt</code>

```
ubuntu:~$ cd app
bash: cd: app: Not a directory
ubuntu:~$ cd data
bash: cd: data: Not a directory
ubuntu:~$ ls
a.txt  app  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ cd a
a.txt  app
ubuntu:~$ cd a
a.txt  app
ubuntu:~$ cd app
bash: cd: app: Not a directory
ubuntu:~$ touch app
ubuntu:~$ ls
a.txt  app  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ mkdir app
mkdir: cannot create directory 'app': File exists
ubuntu:~$ mkdir app2
ubuntu:~$ cd app2
ubuntu:~/app2$ cd ..
ubuntu:~$ rmdir app
rmdir: failed to remove 'app': Not a directory
ubuntu:~$ ls
a.txt  app  app2  b.txt  c.txt  data  filesystem  music  photos  videos
ubuntu:~$ mono app
Command 'mono' not found, did you mean:
  command 'mono' from deb mono-runtime (6.8.0.105+dfsg-3.5ubuntu1)
  command 'nano' from deb nano (7.2-2ubuntu0.1)
  command 'nona' from deb hugin-tools (2023.0.0+dfsg-1)
Try: apt install <deb name>
ubuntu:~$ cd app2
ubuntu:~/app2$ nano file
ubuntu:~/app2$ ls
file
ubuntu:~/app2$ cat file
Hello Linux
This is my first file
ubuntu:~/app2$ vi file
ubuntu:~/app2$ cat file
Hello Linux
This is my first file
ok new line added here ok
```

File Operations

Once you have files, you need commands to view, copy, move, and delete them.

Command	Simple Language Explanation	Example
<code>cat</code>	C atenate. Shows the entire content of a file directly in the terminal.	<code>cat notes.txt</code>
<code>less</code>	Views file content page-by-page, good for large files. Press <code>q</code> to quit.	<code>less very_large_log.txt</code>
<code>cp</code>	C opy. Copies a file from one location to another.	<code>cp file1.txt /home//backup/</code>
<code>mv</code>	M ove. Moves a file (like cutting and pasting). Also used to rename files.	<code>mv oldname.txt newname.txt</code>
<code>rm</code>	R emove. Permanently deletes a file (be careful, there is no Recycle Bin).	<code>rm temp.log</code>
<code>rm -r</code>	Removes an empty or non-empty directory. The <code>-r</code> stands for recursive.	<code>rm -r old_folder</code>

Getting Help

The Linux command line has built-in help features for almost every command.

Command	Simple Language Explanation	Example
<code>man</code>	M anual. Shows the comprehensive reference manual page for a command.	<code>man ls</code>

Command	Simple Language Explanation	Example
<code>--help</code>	A simple option available on most commands that provides a quick list of options.	<code>ls --help</code>

Viewing System Information

These commands provide insights into how your system is running.

Command	Simple Language Explanation	Example
<code>date</code>	Displays the current system date and time.	<code>date</code>
<code>whoami</code>	Tells you the name of the user you are currently logged in as.	<code>whoami</code>
<code>uname -a</code>	Displays detailed information about your operating system kernel.	<code>uname -a</code>
<code>df -h</code>	Disk Free. Shows the amount of available disk space, in "human-readable" format.	<code>df -h</code>
<code>top</code>	Shows a real-time list of running processes, like the Windows Task Manager. Press <code>q</code> to quit.	<code>top</code>

Search and Location

Commands to find files or text within files.

Command	Simple Language Explanation	Example
<code>grep</code>	Global Regular Expression Print. Searches for specific text inside files.	<code>grep "error" server.log</code> (finds the word "error" in the log file)
<code>find</code>	Searches for files and directories based on various criteria (name, size, date).	<code>find . -name "*.txt"</code> (finds all files ending in <code>.txt</code> in the current directory)

Permissions

Linux uses permissions to control who can read, write, or execute a file.

Command	Simple Language Explanation	Example
<code>chmod</code>	Change Mode. Changes the permissions of a file or folder.	<code>chmod +x script.sh</code> (makes a file executable)
<code>chown</code>	Change Owner. Changes the owner of a file or folder.	<code>chown :users important_file.conf</code>

LINUX COMMANDS

1.ls

What: `ls` is a command used to display the list of files and folders in a directory.

Why: It helps you quickly see what contents are available in the current location.

```
controlplane:~$ ls
file1 file2 file3 filesystem folder1 folder2
```

2.touch

What: touch is a command used to create a new empty file or update the timestamp of an existing file.

Why: It helps you quickly make files or refresh file modification time.

```
controlplane:~$ touch file1 file2 file3
```

3.mkdir

What: mkdir is a command used to create a new directory (folder).

Why: It helps you organize files by making separate folders for storage.

```
controlplane:~$ mkdir folder1 folder2
```

4.pwd

What: pwd is a command used to show the full path of the current working directory.

Why: It helps you know exactly where you are in the file system.

```
controlplane:~/folder1$ pwd
/root/folder1
```

5.cd

What: cd is a command used to change the current directory.

Why: It helps you move between folders to access different files and locations.

```
controlplane:~$ cd folder1
controlplane:~/folder1$ cd folder2/folder3
controlplane:~/folder1/folder2/folder3$ cd ..
controlplane:~/folder1/folder2$ cd folder3
controlplane:~/folder1/folder2/folder3$ cd ../../
controlplane:~/folder1$
```

6.vi

What: vi is a command-line text editor used to create and edit files.

Why: It helps you write or modify file content directly from the terminal.

```
controlplane:~$ vi file1
```

- **i ->What:** i is a key in **vi editor** used to enter *Insert mode* for typing text.
Why: It allows you to start editing and adding content inside the file.
- **Esc->What:** Esc is a key in vi used to exit Insert mode and return to Command mode.
Why: It helps you stop typing and run vi commands like save or quit.
- **:wq->What:** :wq is a vi command used to save the file and exit the editor.
Why: It helps you store changes and close vi safely.

7.cat

What: cat is a command used to display the content of a file on the terminal.

Why: It helps you quickly view or check what is inside a file.

```
controlplane:~$ cat file1  
hiiii
```

8.nano

What: nano is a simple command-line text editor used to create and edit files.

Why: It allows easy editing of files directly in the terminal with on-screen instructions.

```
controlplane:~$ nano file1  
controlplane:~$ cat f1  
oohiii
```

changed file content and name to f1

9.rm

What: rm is a command used to delete files or directories.

Why: It helps you remove unwanted files or clean up space.

```
controlplane:~$ ls  
f1 file2 file3 filesystem folder1 folder2  
controlplane:~$ rm file2 file3 folder1  
rm: cannot remove 'folder1': Is a directory  
controlplane:~$ ls  
f1 filesystem folder1 folder2
```

- **Rm -rf <filename>** -> this will delete folder or file even if folder is not empty

```
controlplane:~$ ls
f1 filesystem folder1 folder2
controlplane:~$ rm -rf folder1
controlplane:~$ ls
f1 filesystem folder2
```

10.rmdir

What: rmdir is a command used to delete an empty directory.

Why: It helps you remove unnecessary folders from the system.

```
controlplane:~$ ls
f1 filesystem folder2
controlplane:~$ rmdir folder2
controlplane:~$ ls
f1 filesystem
```

11.history

What: history is a command used to show a list of previously executed commands.

Why: It helps you review or reuse past commands easily.

```
controlplane:~$ history
1  exit
2  halt
3  FILE=/ks/wait-background.sh; while ! test -f ${FILE}; do clear; sleep 0.01; done; bash ${FILE}
4  ls
5  mkdir folder1
6  touch file1
7  history
```

12.cal

What: cal is a command used to display a calendar in the terminal.

Why: It helps you quickly check dates without leaving the command line.

```
controlplane:~$ cal
      January 2026
Su Mo Tu We Th Fr Sa
                1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```

13.df

What: df is a command used to display the available and used disk space on file systems.

Why: It helps you check storage usage and manage disk space.

```
controlplane:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
tmpfs            230020      2568    227452   2% /run
/dev/vda1       19221248 10443404   8761460  55% /
tmpfs           1150084         84   1150000   1% /dev/shm
tmpfs            5120         0       5120   0% /run/lock
/dev/vda16       901520    119048   719344  15% /boot
/dev/vda15       106832      6250    100582   6% /boot/efi
tmpfs           230016         96    229920   1% /run/user/112
shm             65536         0     65536   0% /run/containerd/io.containerd.grpc.v1.cri/sandboxes/77d577263693bd800
fde271156bf162ec69379f07e0d951d0c0f705496d46441/shm
shm             65536         0     65536   0% /run/containerd/io.containerd.grpc.v1.cri/sandboxes/c8589c56b457447fc
24285c71e7bea78e560dd3e76b2958b470ef04a31e34a72/shm
shm             65536         0     65536   0% /run/containerd/io.containerd.grpc.v1.cri/sandboxes/f3b4bf9e9ee7fe62
17a0f23e19d6f3431f891b0b3fb1a1be843db768b72a6d8/shm
shm             65536         0     65536   0% /run/containerd/io.containerd.grpc.v1.cri/sandboxes/5710489d7e89dc4a9
7ced2b10d6e8b2250830fc013d7db41a96d09f2fc5ced56/shm
overlay         19221248 10443404   8761460  55% /run/containerd/io.containerd.runtime.v2.task/k8s.io/77d577263693bd80
0fde271156bf162ec69379f07e0d951d0c0f705496d46441/rootfs
```

14.whoami

What: whoami is a command used to display the current logged-in user.

Why: It helps you confirm which user account is active in the session.

```
controlplane:~$ whoami
root
```

15.ping

What: ping is a command used to check the network connectivity to a server or IP address.

Why: It helps you verify if a device or website is reachable and measure response time.

```
controlplane:~$ ping www.google.com
PING www.google.com (172.217.20.36) 56(84) bytes of data.
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=1 ttl=115 time=12.9 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=2 ttl=115 time=13.2 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=3 ttl=115 time=13.1 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=4 ttl=115 time=12.7 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=5 ttl=115 time=12.5 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=6 ttl=115 time=12.7 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=7 ttl=115 time=12.8 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=8 ttl=115 time=12.7 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=9 ttl=115 time=12.9 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=10 ttl=115 time=13.1 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=11 ttl=115 time=12.8 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=12 ttl=115 time=13.0 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=13 ttl=115 time=12.9 ms
64 bytes from arn11s01-in-f4.1e100.net (172.217.20.36): icmp_seq=14 ttl=115 time=12.8 ms
```

16.mv

What: mv is a command used to move or rename files and directories.

Why: It helps you organize files by relocating them or changing their names.

```
controlplane:~$ mv file1 omkar1
controlplane:~$ ls
filesystem  folder1  omkar1
```



```
controlplane:~$ ls
filesystem folder1 omkar1
controlplane:~$ mv omkar1 folder1
controlplane:~$ ls
filesystem folder1
controlplane:~$ cd folder1
controlplane:~/folder1$ ls
omkar1
```

17.cp

What: cp is a command used to copy files or directories.

Why: It helps you create duplicates for backup or use in different locations.

```
controlplane:~/folder1$ cp omkar1 omkar2
controlplane:~/folder1$ ls
omkar1 omkar2
```

18.wget

What: wget is a command used to download files from the internet using a URL.

Why: It helps you quickly fetch web content or software directly from the terminal.

```
controlplane:~/folder1$ wget https://downloads.apache.org/tomcat/tomcat-11/v11.0.18/bin/apache-tomcat-11.0.18.exe
--2026-01-30 06:33:58-- https://downloads.apache.org/tomcat/tomcat-11/v11.0.18/bin/apache-tomcat-11.0.18.exe
Resolving downloads.apache.org (downloads.apache.org)... 88.99.208.237, 135.181.214.104, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|88.99.208.237|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14817720 (14M) [application/x-msdos-program]
Saving to: 'apache-tomcat-11.0.18.exe'

apache-tomcat-11.0.18.exe 100%[=====>] 14.13M 6.36MB/s in 2.2s

2026-01-30 06:34:01 (6.36 MB/s) - 'apache-tomcat-11.0.18.exe' saved [14817720/14817720]
```

Linux File Permission

✓ `ls -l`

Shows file and directory permissions in long format.

```
ubuntu:~$ ls -l
total 0
lrwxrwxrwx 1 root root 1 Jan 21 14:56 filesystem -> /
ubuntu:~$
```

✓ `ls -ld directory_name`

Shows permissions of the directory itself (not its contents).

```
ubuntu:~$ ls -ld dip
drwxr-xr-x 2 root root 4096 Jan 30 09:16 dip
ubuntu:~$
```

```
ubuntu:~$ ls
dip dir1~ file1.txt file2.txt filesystem script.sh
ubuntu:~$ stat file1.txt
  File: file1.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 253,1    Inode: 7749          Links: 1
Access: (0644/-rw-r--r--)  Uid: (  0/   root)   Gid: (  0/   root)
Access: 2026-01-30 09:21:50.333462189 +0000
Modify: 2026-01-30 09:21:50.333462189 +0000
Change: 2026-01-30 09:21:50.333462189 +0000
 Birth: 2026-01-30 09:21:50.333462189 +0000
ubuntu:~$
```

✓ `chmod 755 file`

Changes file permissions (read/write/execute) using numeric mode.

```
ubuntu:~$ chmod 644 file1.txt
ubuntu:~$ chmod 600 file2.txt
ubuntu:~$ chmod 755 script.sh
ubuntu:~$ ls
dip dir1~ file1.txt file2.txt filesystem script.sh
ubuntu:~$
```

✓ chmod u+x file

Adds execute permission to the file owner.

```
dip dir1~ file1.txt file2.txt filesystem script.sh
ubuntu:~$ ./script.sh
./script.sh: line 1: hello: command not found
ubuntu:~$ ./script.sh
./script.sh: line 1: hello: command not found
ubuntu:~$ echo "echo hello dip" > script.sh
ubuntu:~$ cat script.sh
echo hello dip
ubuntu:~$ ./script.sh
hello dip
ubuntu:~$
```

✓ chmod g+w file

Adds write permission to the group.

✓ chmod o-r file

Removes read permission from others.

✓ **chmod a+x file**

Adds execute permission for all users.

✓ **chmod -R 755 folder**

Changes permissions recursively for a directory and its files.

✓ **chown user file**

Changes the owner of a file.

✓ **chown user:group file**

Changes both owner and group of a file.

✓ **chgrp group file**

Changes only the group of a file.

✓ **stat file**

Shows detailed information about a file including permissions, owner and timestamps.

✓ **getfacl file**

Displays Access Control List (ACL) permissions for a file.

✓ **setfacl -m u:user:rw file**

Sets extra ACL permissions for a specific user on a file.

File & Directory Commands

Command	Use
<code>ls</code>	List files and folders
<code>pwd</code>	Show current directory
<code>cd</code>	Change directory
<code>mkdir</code>	Create a directory
<code>rmdir</code>	Delete empty directory
<code>touch</code>	Create empty file
<code>cp</code>	Copy files / folders
<code>mv</code>	Move or rename files
<code>rm</code>	Delete files / folders
<code>tree</code>	Show folder structure

```
ubuntu:~$ tree
.
|-- dip
|   |-- data2
|   |-- hello
|   |-- photos
|   `-- videos
|-- filesystem -> /
|-- games
|-- movies
`-- music
```

```
ubuntu:~$ ls
filesystem
ubuntu:~$ mkdir games music movies dip
ubuntu:~$ ls
dip filesystem games movies music
ubuntu:~$ cd dip
ubuntu:~/dip$ mkdir photos videos
ubuntu:~/dip$ mkdir data
ubuntu:~/dip$ ls
data photos videos
ubuntu:~/dip$ mv data data2
ubuntu:~/dip$ ls
data2 photos videos
ubuntu:~/dip$ touch hello
ubuntu:~/dip$ ls
data2 hello photos videos
ubuntu:~/dip$ rm data2
rm: cannot remove 'data2': Is a directory
ubuntu:~/dip$ ls
data2 hello photos videos
ubuntu:~/dip$ vi hello
ubuntu:~/dip$ cat hello
This is the data in the file
more written here
ubuntu:~/dip$ head hello
This is the data in the file
more written here
ubuntu:~/dip$ more hello
This is the data in the file
more written here
ubuntu:~/dip$ cd ..
```

File View & Edit

Command	Use
<code>cat</code>	View file content
<code>less</code>	View large file page by page

<code>more</code>	Simple file viewer
<code>head</code>	First lines of file
<code>tail</code>	Last lines of file
<code>tail -f</code>	Live log view
<code>nano</code>	Simple editor
<code>vi / vim</code>	Advanced editor

```
ubuntu:~/dip$ vi hello
ubuntu:~/dip$ cat hello
This is the data in the file
more written here
ubuntu:~/dip$ head hello
This is the data in the file
more written here
ubuntu:~/dip$ more hello
This is the data in the file
more written here
```

Search & Find

Command	Use
<code>find</code>	Find files
<code>locate</code>	Fast file search
<code>grep</code>	Search text in files
<code>which</code>	Command location
<code>whereis</code>	Binary / source location

```
data2 hello photos videos
ubuntu:~/dip$ cat hello
This is the data in the file
more written here
ubuntu:~/dip$ grep "data" hello
This is the data in the file
ubuntu:~/dip$
```

```
ubuntu:~$ whereis python
python: /usr/bin/python
ubuntu:~$ whereis java
java: /usr/share/java
ubuntu:~$
```

File Information

Command	Use
<code>file</code>	File type
<code>stat</code>	File details
<code>du</code>	File/folder size
<code>df</code>	Disk usage

```
data2 hello photos videos
ubuntu:~/dip$ file hello
hello: ASCII text
ubuntu:~/dip$ stat hello
  File: hello
  Size: 47          Blocks: 8          IO Block: 4096   regular file
Device: 253,1  Inode: 561441      Links: 1
Access: (0644/-rw-r--r--)  Uid: (  0/   root)   Gid: (  0/   root)
Access: 2026-01-31 04:55:47.769602119 +0000
Modify: 2026-01-31 04:55:42.160571601 +0000
Change: 2026-01-31 04:55:42.177571694 +0000
  Birth: 2026-01-31 04:55:42.160571601 +0000
ubuntu:~/dip$
```



```

data2 hello photos videos
ubuntu:~/dip$ file hello
hello: ASCII text
ubuntu:~/dip$ stat hello
  File: hello
  Size: 47          Blocks: 8          IO Block: 4096   regular file
Device: 253,1    Inode: 561441      Links: 1
Access: (0644/-rw-r--r--)  Uid: (   0/   root)   Gid: (   0/   root)
Access: 2026-01-31 04:55:47.769602119 +0000
Modify: 2026-01-31 04:55:42.160571601 +0000
Change: 2026-01-31 04:55:42.177571694 +0000
 Birth: 2026-01-31 04:55:42.160571601 +0000
ubuntu:~/dip$ du data2/
4      data2/
ubuntu:~/dip$ du hello
4      hello
ubuntu:~/dip$ df photos/
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/vda1        19221248 5419492  13785372  29% /
ubuntu:~/dip$ du hello
4      hello
ubuntu:~/dip$ █

```

User & Permission Commands

Command	Use
<code>whoami</code>	Current user
<code>id</code>	User info
<code>su</code>	Switch user
<code>sudo</code>	Run as admin
<code>chmod</code>	Change permission
<code>chown</code>	Change owner
<code>chgrp</code>	Change group

```

ubuntu:~$ whoami
root
ubuntu:~$ id
uid=0(root) gid=0(root) groups=0(root)
ubuntu:~$ us
us: command not found
ubuntu:~$ su
ubuntu:~$ sudo useradd dip
useradd: group dip exists - if you want to add this user to that group, use -g.
ubuntu:~$ sudo useradd diphire
ubuntu:~$ sudo useradd diphire2
ubuntu:~$ sudo adduser diphire3
info: Adding user `diphire3' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `diphire3' (1003) ...
info: Adding new user `diphire3' (1003) with group `diphire3 (1003)' ...
info: Creating home directory `/home/diphire3' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for diphire3
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `diphire3' to supplemental / extra groups `users' ...
info: Adding user `diphire3' to group `users' ...
ubuntu:~$ su diphire
\[\e]0;\h: \w\a\]\h:\w$ █

```

User Management (Important for admin / DevOps)

Command	Use
<code>useradd</code>	Create user
<code>usermod</code>	Modify user
<code>userdel</code>	Delete user

groupadd	Create group
groupdel	Delete group
passwd	Set password

```
ubuntu:~$ su
ubuntu:~$ sudo useradd dip
useradd: group dip exists - if you want to add this user to that group, use -g.
ubuntu:~$ sudo useradd diphire
ubuntu:~$ sudo useradd diphire2
ubuntu:~$ sudo adduser diphire3
info: Adding user `diphire3' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `diphire3' (1003) ...
info: Adding new user `diphire3' (1003) with group `diphire3 (1003)' ...
info: Creating home directory `/home/diphire3' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for diphire3
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
info: Adding new user `diphire3' to supplemental / extra groups `users' ...
info: Adding user `diphire3' to group `users' ...
ubuntu:~$ su diphire
\[e]0;\h: \w\a\]\h:\w$ sudo userdel diphire2
[sudo] password for diphire:
```

Process & System Monitoring

Command	Use
ps	Show processes

<code>top</code>	Live process view
<code>htop</code>	Better top (if installed)
<code>kill</code>	Kill process
<code>killall</code>	Kill by name
<code>uptime</code>	System running time
<code>free</code>	RAM usage
<code>vmstat</code>	Memory stats

```
ubuntu:~$ ps
  PID TTY          TIME CMD
  3420 pts/3    00:00:00 bash
  3440 pts/3    00:00:00 ps
ubuntu:~$ htop
ubuntu:~$ kill
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
ubuntu:~$ free
             total        used        free      shared  buff/cache   available
Mem:      1948940       499264       438908         1092     1208460     1449676
Swap:      1048572           0       1048572
ubuntu:~$ vmstat
procs -----memory-----  ---swap--  -----io----  -system--  -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi   bo    in   cs us  sy id  wa st gu
  2  0        0 438684 126688 1081808    0    0   152   105  103    2  1  0 99  0  0  0
ubuntu:~$
```

Networking

Command	Use
<code>ip a</code>	Network interfaces
<code>ping</code>	Network check
<code>ss</code>	Port & socket info
<code>netstat</code>	Old port command

curl

API / URL request

wget

Download files

```
ubuntu:~$ ping google.com
PING google.com (74.125.68.113) 56(84) bytes of data:
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=1 ttl=101 time=1.48 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=2 ttl=101 time=1.22 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=3 ttl=101 time=1.23 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=4 ttl=104 time=1.40 ms
64 bytes from sc-in-f113.1e100.net (74.125.68.113): icmp_seq=5 ttl=104 time=1.33 ms
^Z
[1]+  Stopped                  ping google.com
ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 2e:c2:cb:8f:f0:4e brd ff:ff:ff:ff:ff:ff
    inet 172.30.1.2/24 brd 172.30.1.255 scope global dynamic noprefixroute enp1s0
        valid_lft 86308075sec preferred_lft 75518875sec
    inet6 fe80::41b7:ed1:dc49:df5a/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1454 qdisc noqueue state DOWN group default
    link/ether 06:a3:79:0d:2b:73 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
ubuntu:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 172.30.1.2:40200        10.244.3.246:46002     ESTABLISHED
tcp        0      0 172.30.1.2:40205        10.244.4.187:55310     ESTABLISHED
tcp        0      0 172.30.1.2:40200        10.244.3.252:58036     ESTABLISHED
tcp        0      0 172.30.1.2:40200        10.244.3.252:33694     ESTABLISHED
```



Package Management (Ubuntu)

Command	Use
apt update	Update repo
apt upgrade	Upgrade packages
apt install	Install software

`apt remove`

Remove
software



Compression & Archive

Command	Use
<code>tar</code>	Create / extract archive
<code>gzip</code>	Compress
<code>gunzip</code>	Decompress
<code>zip</code>	Zip file
<code>unzip</code>	Unzip file

```
ubuntu:~/dip$ gzip hello
ubuntu:~/dip$ ls
data2 hello.gz photos videos
ubuntu:~/dip$ zip hello.gz
Command 'zip' not found, but can be installed with:
apt install zip
ubuntu:~/dip$ unzip hello.gz
Archive:  hello.gz
  End-of-central-directory signature not found.  Either this file is not
  a zipfile, or it constitutes one disk of a multi-part archive.  In the
  latter case the central directory and zipfile comment will be found on
  the last disk(s) of this archive.
unzip:  cannot find zipfile directory in one of hello.gz or
        hello.gz.zip, and cannot find hello.gz.ZIP, period.
ubuntu:~/dip$
```



Redirection & Pipes (VERY important)

Symbol	Use
<code>></code>	Write output to file

<code>>></code>	Append output
<code><</code>	Input from file
<code>,</code>	,

System & OS Info

Command	Use
<code>uname -a</code>	Kernel info
<code>hostname</code>	System name
<code>hostnamectl</code>	Host info
<code>lsb_release -a</code>	OS info
<code>date</code>	Date and time
<code>cal</code>	Calendar

Mount & Storage

Command	Use
<code>mount</code>	Mount disk
<code>umount</code>	Unmount
<code>lsblk</code>	Disk layout
<code>blkid</code>	Disk UUID

SSH & Remote

Command	Use
ssh	Remote login
scp	Copy files remote
rsync	Fast sync